

SELECTED PROJECTS · ENGINEERING DOSSIER

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A curated dossier of production systems, deployed agentic-AI demos, and engineering clusters — each built solo, end-to-end. Flagships first, then live AI demos, then the subsystem clusters behind them.

FLAGSHIP SYSTEMS

FLAGSHIP

Instantino

B2B SaaS in production — the deep proof.

A full B2B SaaS built solo, end-to-end: turns a business URL into a live, multilingual quote form plus an instant branded PDF offer. Multi-tenant PHP + MySQL, 6 languages, ~1,800 programmatic SEO pages, a conversational form editor, and a parametric pricing engine. Its subsystems populate the clusters below.

STACK

PHP 8 MySQL React Tailwind Stripe mPDF Claude API GPT API

HIGHLIGHTS

- Quickstart: URL → AI niche detection → full parametrically-priced quote form + real PDF
- ~1,800 programmatic SEO pages auto-built from the live product catalog
- 6-language support with incremental translation memory
- Conversational form editor with money-guarded price edits
- Anonymous priced-PDF demo that back-solves a cost breakdown from one price
- Economic-plausibility simulation as an anti-hallucination guard on AI prices

Live: <https://instantino.ro>

FLAGSHIP

Orchestration Platform

Multi-agent AI framework — proven deep and wide.

A self-hosted multi-tenant control plane for AI coding agents: token-based authorization, scope-jailing, project scaffolding, and a self-improving lessons loop. Proven two ways: DEEP (single-project — Instantino, ~17 systems) and WIDE (multi-project / multi-user deployment with per-project scoped tokens). The framework built itself: 27 briefs → 26 structured reports. Its live-ops layer, Nexus Cockpit, monitors the whole stack from a terminal dashboard.

STACK

PHP 8 MCP (stdio + HTTP) JSON-RPC 2.0 Claude Code CLI PowerShell Python + Textual SQLite

HIGHLIGHTS

- Token-based RBAC: master vs per-project scoped tokens with expiry/revocation
- Scope-jailing: project tokens cannot read sibling projects
- Project scaffolder: scaffold/integrate/archive with 9 auto-rendered templates
- Self-building: 27-brief dependency-ordered build program, 26 completed reports
- Dual transport: stdio (local) + HTTP :8090 (JSON-RPC 2.0)
- AI-learning loop: detect gotcha → draft lesson → owner approves → auto-injected into every session
- Nexus Cockpit — Python/Textual live-ops dashboard (9 panels) logging every MCP / CLI / Guardian call to SQLite

FLAGSHIP

AI Support Console (iPDF Live)

Human + AI customer support that learns from every escalation.

A production AI customer-support console. Live conversations arrive from multiple surfaces (public site, client panel, quick-start); the AI drafts a grounded reply for the human agent, and anything it can't confidently answer is escalated to a person — behind a TOTP-secured login. An Intelligence Center tracks conversations, unique sessions and off-topic rate, and surfaces Knowledge Gaps — the top questions the AI couldn't answer. Agents can teach the correct answer on the spot; it is saved to a RAG store and reused, so the system gets smarter with every escalation. A built-in Test Engine sandbox probes the AI's behavior per source and language before anything ships.

STACK

PHP 8 MySQL vanilla JS Claude / GPT API RAG TOTP 2FA

HIGHLIGHTS

- Live multi-source inbox (public site / client panel / quick-start), multilingual, polling-based
- AI drafts a grounded reply for the human agent — human-in-the-loop, never autonomous
- Escalation to a person for anything the AI can't confidently answer
- Intelligence Center: conversations, unique sessions, off-topic rate + Knowledge Gaps (top unanswered)
- Self-improving RAG: teach the correct answer once → saved → reused by the AI
- Test Engine sandbox to probe AI behavior per source + language before go-live
- TOTP-secured agent login

LIVE AI DEMOS

DEPLOYED DEMO

Taktile — AI Enablement Console

Governed multi-agent AI over a real MCP server.

A deployed demo of governed agentic AI: an AI agent wires a company's SaaS stack (Gong, Notion, Linear, HubSpot) into real internal workflows — but the model only proposes; the server decides. Every action runs under a least-privilege scope and is written to an append-only audit trail, so the agent acts across real tools without ever overstepping. Not a website — a working agentic system with a real MCP server and an eval suite, shipped end-to-end to a Linux VPS.

STACK

Python Claude (Haiku) MCP server FastAPI evals Linux VPS

HIGHLIGHTS

- The model proposes, the server decides — server-side least-privilege enforcement on every tool call
- Append-only audit trail: timestamp, request id, actor, model, tool, scope, decision
- Real MCP server (tools/list + tools/call) — plugs straight into Claude Desktop
- Eval suite asserting the governance invariants (a denied action never executes)
- Grounding post-check flags any id not traced to an allowed tool result
- Shipped end-to-end: systemd, Apache reverse proxy, HTTPS, rate-limiting

Live: <https://taktile.instantino-saas.eu/>

DEPLOYED DEMO

Netex — AI Front-Line Triage

Grounded multilingual triage — the model classifies, the server decides.

A deployed AI front-line for inbound customer messages: it detects the language, classifies intent / urgency / sentiment, drafts a reply grounded strictly in a knowledge base, and decides — by a deterministic, server-side rule engine — whether to auto-reply or escalate to a human. Prompt-injection is detected and blocked

server-side, structured JSON outputs are schema-enforced, and QA checks guard every draft. Not a website — a working AI system where a business decision never rests on the model alone.

STACK

PHP 8 gpt-5-nano structured outputs rule engine KB grounding

HIGHLIGHTS

- Deterministic server-side escalation rules — the model classifies, the server decides
- Grounded strictly in a knowledge base; if the answer isn't there, it escalates instead of guessing
- Prompt-injection detected and blocked server-side (not left to the model)
- Schema-enforced structured JSON outputs with a json_object fallback
- Binary QA checks + language-match guard on every draft
- Multilingual + multichannel (Webchat / WhatsApp / Facebook / Voice / Email)

Live: <https://netex.instantino-app.eu/>

ENGINEERING CLUSTERS

CLUSTER

AI Forms & Pricing

From URL to priced PDF — the original product chain.

The novel core of Instantino: an AI-driven pipeline that takes a business URL, detects its niche, generates a parametrically-priced multilingual form, and produces a real branded PDF quote — with preview-before-signup and human-in-the-loop approval. Includes a declarative pricing DSL, a no-code form pipeline that emits real PHP, and an embedded AI copilot with a money-guarded editor.

STACK

PHP 8 Claude API GPT API mPDF Tailwind

HIGHLIGHTS

- URL → AI niche → parametric pricing model → multilingual form → real PDF
- Economic-plausibility simulation guards AI-suggested prices
- Quality-driven model escalation: validator triggers Sonnet → Opus
- Money-guarded conversational form editor (confirm widget on price edits)
- No-code pipeline emits real deployable PHP, not hosted/interpreted
- Human-in-the-loop queue curates AI-generated forms into the catalog

CLUSTER

Lead-gen & Outreach

Built 4 times, each more sophisticated.

Four progressively more advanced lead-generation and cold-outreach systems: from AI discovery and email generation to automatic SMTP warm-up, a sales-engagement command center, and a standalone deployed web app. Each iteration refined deliverability engineering, reply handling, and AI personalization.

STACK

PHP 8 MySQL SMTP IMAP Serper API GPT API Chart.js

HIGHLIGHTS

- AI discovery → crawl → extract → validate → AI email → SMTP → IMAP monitoring
- 160-line email validator + RFC-8058 one-click unsubscribe
- Automatic SMTP warm-up + account testing (DNS/RBL/Postmark)
- AI reply grounded in the lead's own form schema
- Sales-engagement command center: 6 sub-tabs, follow-up queue, email-health
- ~11k leads discovered, ~8.6k emails sent across iterations

CLUSTER

SEO · Analytics · Security

Privacy analytics, programmatic SEO, and security — all built from scratch.

A suite covering visibility and safety: a programmatic SEO engine that auto-builds ~1,800 pages from the live product catalog, a 2-engine SEO auditor strong on hreflang/i18n/schema, cookieless privacy analytics with dark-funnel re-identification, and a regex-based security SAST scanner with auto-fix.

STACK

PHP 8 MySQL GPT API geo-IP

HIGHLIGHTS

- ~1,800 programmatic SEO pages from 5 intent layers × 60 niches × 6 langs
- Self-rebuilding registry: add a niche → entire SEO surface regenerates
- Cookieless analytics: daily-salted SHA-256 IP hash, no raw IP stored
- Dark-funnel bridge: re-identifies anonymous visitors as named CRM leads
- 2-engine SEO auditor strong on hreflang / i18n parity / schema
- Regex SAST: 18 OWASP patterns + active HTTP probes + auto-fix

CLUSTER

Business & Growth Ops

The systems that run and grow the business.

Cross-cutting business and growth systems, each built from scratch and mapping to a paid SaaS category: an AI multilingual content CMS, a social auto-publisher (Facebook + LinkedIn), incremental translation memory, a full Stripe subscription lifecycle, a coupon/campaign engine, and a clients operations grid.

STACK

PHP 8 MySQL Stripe GPT API Facebook Graph API LinkedIn API

HIGHLIGHTS

- AI multilingual content CMS with anti-hallucination grounding
- Social auto-publish to Facebook (Graph) + LinkedIn (auto token-refresh)
- Incremental translation memory — translate only changed keys + coherence check
- Full Stripe subscription lifecycle: plans × regions, 7 webhook events, top-ups
- Coupon/campaign engine with duration tiers + Stripe-coupon bridge
- Clients ops grid: inline pricing/SMTP edit, embed-token regen, plan-grant

CLUSTER

Apps & Internal Tools

A shipped product + an internal enterprise portal.

Audisity — a packaged Windows file-audit product with offline licensing, in real-world use; and MailDesk — an internal IT portal for email-account provisioning, in daily production on enterprise integrations.

STACK

Python 3 PySide6 (Qt6) PyInstaller PHP 8 Zimbra SOAP RT REST

HIGHLIGHTS

- Audisity — Windows file-audit (ZIP/JPG/PST/age) → XLSX, HMAC offline licensing; in real-world use + built for commercial sale
- Thread-safe Qt async UI (8 worker threads) + single-exe distribution
- MailDesk — internal IT portal: Zimbra account provisioning via RT ticketing, in daily production
- Built on enterprise integrations: Zimbra Admin SOAP + Request Tracker REST

CLUSTER

Design & Frontend

Design range — from luxury hotel to transport logistics.

Frontend and design showcases that balance the heavy-backend projects: a luxury hotel booking landing, a bilingual transport company site with embedded Instantino forms, and a lead-gen dashboard UI. Pure design and frontend craft.

STACK

HTML5 CSS3 Tailwind vanilla JS Chart.js

HIGHLIGHTS

- Responsive layouts with CSS Grid, Flexbox, clamp() typography
- IntersectionObserver scroll-reveal animations
- Canvas radial patterns and animated counters
- Embedded Instantino form integration (real-world usage)

DESIGN PIECES

Maison Elegante

Luxury hotel booking landing — responsive single-page design showcase with Playfair + Poppins typography.

Live: <https://maison-elegante.instantino-app.eu/>

Carpatica Transport

Bilingual (RO/EN) marketing site for a heavy/oversized-cargo transport company. Embeds an Instantino quote form — real-world integration.

Live: <https://carpatica.instantino.ro>

Scrapix Dashboard

Lead-gen + outreach dashboard UI with Chart.js analytics, 7 tabs, deployed at affizyletter.com.

Client Portfolio (Sondos)

A personal portfolio site built for a client — responsive single-page design.

Live: <https://portfolio.try-instantino.eu/>
